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CHM 101 — Past Questions & Materials

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Course: CHM 101

Institution: Osun State University, Osogbo

Papers in this pack: 7

Levels: 100

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Paper 1 of 7 — CHM 101 GENERAL CHEMISTRY I

Level 100 • First Semester • Handout

CHM 101

GENERAL CHEMISTRY I

COURSE LECTURER: Hassan K. Busari

1

COURSE SYNOPSIS

> Chemical Bonding: Electrovalent bond between ions.

Covalent bonds. Coordinate covalent bond. Metallic bonds.

Intermolecular forces. Hydrogen bonding and its influence on properties. Kinetic theory of matter

2

CHEMICAL BONDING

IONIC BONDING: involves transfer of electrons from metal (forming positive ion) to non-metal (forming negative

ion). Example

Na X

X X

Na

. X X

Cl X X

Cl X X X

X X X X

1s²2s²2p⁶3s¹ 1s²2s²2p⁶3s²3p⁵

1s²2s²2p⁶ 1s²2s²2p⁶3s²3p⁶

Na loses an electron while Cl gains an electron to attain octet configuration. Other examples: MgCl₂, Li₂O

COVALENT BONDING: involves sharing of electrons. Other examples: O₂, H₂, N₂

. XX .

. X X

.. C

I XIX

CX .

..Cl C IX X

. X

X .

. X

X X

Chlorine molecule (Cl₂)

Coordinate Covalent Bonding: the shared pair of electrons is donated by only one of the bonded atoms

H

.. H

. . .

H.N H.N

+ H

.. H

H H 3

Properties of ionic compounds

- Solid with high melting and boiling points
- Conduct electricity when molten, but not as a solid
- It is made from a combination of metal and non-metal elements

Properties of covalent compounds

- Low boiling/melting points
- Soft in nature and relatively flexible
- Poor conductors of heat and electricity
- It is made from combination of non-metal elements

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Metallic bonding: is the electrostatic attraction between the delocalized electron cloud and metal ions. It holds

metal ion tightly and confers high melting point.

Factors affecting metallic bonding

- > Size of the metal ion: it increases with decrease in size of metal ion
- > Number of valence electrons: it increases with increase in number of valence electrons

Intermolecular forces: two types – van der Waal's forces and hydrogen bonding. Both describe electrostatic

attraction between dipoles i.e. the attraction between the positive end of one molecule and negative end of

another molecule

Types of Dipoles:

Permanent dipole: it exists in all polar molecules due to difference in electronegativity

Instantaneous dipole: it occurs as a result of fluctuations in the electron cloud

Temporary

Induced dipole: is created due to influence of a neighbouring dipole

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Van der Waal's: includes dipole-dipole, dipole-induced dipole and instantaneous dipole-induced dipole

interactions

Dipole – dipole interactions: it occurs when polar molecules orientate themselves in order to maximise attractive

forces between molecules while repulsive forces are minimized

Dipole - induced dipole: when a non-polar molecule approaches a polar molecule, a dipole will be induced in the

non-polar molecule in d u c e d d i p o l e i n n o n - p o l a r
m o l e c u l e

p e r m a n e n t d i p o l e

Instantaneous dipole - induced dipole: it occurs from fluctuations of electron cloud and can induce dipole moment

in neighbouring atoms. The greater the number of electrons in a molecule, the easier it is for instantaneous dipole to

be set up. The force of attraction between two temporary dipoles is known as a London force or dispersion force

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Hydrogen bonding: it exists when hydrogen is bonded to highly electronegative atom. It is the strong dipole-dipole interaction between hydrogen and electronegative atom (F, O, Cl, N). It is weaker than covalent bond but stronger than van der Waal's forces. For hydrogen bonding to occur

- A hydrogen atom must be bonded directly to highly electronegative atom
- An unshared pair of electron on the electronegative atom

Hydrogen bonds are responsible for:

- (i) The relatively high melting and boiling points of water and hydrogen fluoride
- (ii) Holding the strands of DNA together 7

Intermolecular and intramolecular hydrogen
bonding

Hydrogen bonds that occur between two different molecules are called intermolecular hydrogen bonds e.g H_2O and NH_3 . Hydrogen bonds that occur within the same molecule are called intramolecular hydrogen bonds e.g 2-Nitrophenol.

- (a) Intermolecular hydrogen bonds (b) Intramolecular hydrogen bond

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KINETIC THEORY OF MATTER

- 1) All matter is composed of atoms and molecules that are constantly in motion
- 2) The state of matter depends on the motion and arrangement of these particles, which is determined by temperature-higher temperatures mean higher particle motion
- 3) The three states of matter-solids, liquids and gases-can be distinguished based on how close together and how freely the particles can move

Note:

- At the same temperature, the heavier particles move slower than the

lighter particles.

- Temperature is a measure of average kinetic energy (K.E) in a molecule

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Solids, liquids & gases

Solids

- Solids have a fixed volume and shape and they have a high density.
- The atoms vibrate in position but can't change location
- The particles are packed very closely together in a fixed and regular pattern

Liquids

- Liquids also have a fixed volume but adopt the shape of the container
- They are generally less dense than solids (an exception is water), but much denser than gases
- The particles move and slide past each other which is why liquids adopt the shape of the container and also why they are able to flow freely

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Solids, liquids & gases

Gases

- Gases do not have a fixed volume, and, like liquids, take up the shape of the container (Indefinite shape and volume)
- Gases have a very low density
- Since there is a lot of space between the particles, gases can be compressed into a much smaller volume
- The particles are far apart and move randomly and quickly (around 500 m/s) in all directions
- They collide with each other and with the sides of the container (this is how pressure is created inside a can of gas)

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Summary of the properties of solids, liquids and gases

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Phase Changes

- Melting/Freezing
- Boiling (vaporization)/Condensing
- Sublimation
- Evaporation

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Melting/Freezing Point

- Change from solid to liquid and liquid to solid.
- Same temperature.; if melting, particles are gaining energy; if

freezing, particles are losing energy.

- The stronger the intermolecular forces (IF's), the more energy needed to weaken the IF's, therefore higher melting point temperature.

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Melting/Freezing Continued

- During the phase change, the temp. remains constant
- After all the sample has changed phase, the temp. will change.
- During the phase change, potential energy (P.E.) is changing, but K.E. is constant.

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Boiling (Vaporization)/Condensation Point

- Change from liquid to gas and gas to liquid.
- Same temp.; if boiling, particles are gaining energy; if condensing, particles are losing energy
- The stronger the IF's, the more energy needed to break the IF's, therefore higher boiling point temperature

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Boiling (Vaporization)/Condensation Point

- During the phase change, the temp. remains constant
- After all the sample has changed phase, the temp. will change.
- During the phase change, potential energy (P.E.) is changing, but K.E. is constant.

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Sublimation

- Changing directly from a solid to a gas.
- Also, changing directly from a gas to a solid
- Skipping the liquid state.

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Evaporation

- Liquid to gas but not necessarily at the boiling point temperature.
- Some particles gain enough K.E. to overcome the IF's and become a gas.
- Remember, temperature is a measure of the average K.E.

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Reading List:

➤ Abass A. Olajire. Fundamental University Organic and Inorganic Chemistry. Sina 2tees publication. 2008.

> Sachin Kr. Ghosh. Advanced General Organic Chemistry (A modern Approach). New Central Book Agency (P)

Limited. 2007

> Wong Y.C., Wong C.T., Onyinruka S.O., Akpanisi L.E. University General Chemistry Inorganic and Physical. Africana-

FEP Publishers LTD

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Paper 2 of 7 — CHM 101 power point 2025

Level 100 • First Semester • Handout

CHM 101

ACIDS, BASES AND SALTS.

REDOX REACTIONS

BY

MR ADENIYI TAYO

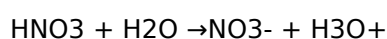
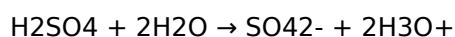
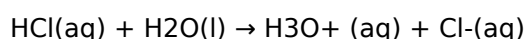
- Acids generally have a sour taste and react strongly with a wide variety of substances such as metals.
- Bases are known to react with acids.
- Early chemists found it difficult to decide a reason why acids had a set of properties in common
- According to the idea from a French chemist “Antoine Lavoisier” he described “acids as substances that contain oxygen”
- This was valid for a number of common acids such as sulphuric acid [H₂SO₄] and nitric acid [HNO₃] but does not fit hydrochloric acid [HCl]
- The search for common factor in acids came to a head during 1830’s when a german Chemist “Justus Von Liebig” wrote that “Acids are hydrogen compounds in which the hydrogen may be replaced by metals.
- All the work on acids has led to the following standard definition of an acid

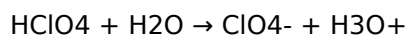
Acid will;

- Give out hydrogen with a metal; a derivative of salt
- Neutralize a base to give a salt and water only
- Give carbon-dioxide with carbonate
- This summary gives a great information about the properties of acids but does not tell us anything about the chemical structure except that they contain hydrogen as one of their elements.
- In the study of acids; it was noted that when acids was found in solutions, it conducted electricity extremely well. This experimental evidence tells us that acid contains ions
- “Acids give hydrogen ions in solution”
- When a molecule like hydrogen chloride dissolves in water, it dissociates into ions;
- $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}^+ (\text{aq}) + \text{Cl}^- (\text{aq})$

Experimental evidence showed that the active ingredient of an acidic solution is not a simple hydrogen ion, H⁺ (aq) rather, it is the oxonium ion, H₃O⁺ (aq) {oxonium ions are also known as hydronium ion}

Hence when hydrogen chloride dissolves in water, there is a chemical reaction;





{Chloric (vii) acid}

The Bronsted theory of acids and bases

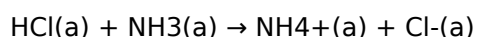
Due to the restriction of the previous definition of acids, bronsted brought out his own theory having noted that some chemist carry out reaction in non-aqueous solvent such as liquid ammonia and liquid sulphur dioxide.

He stated that;

Acids are proton donors

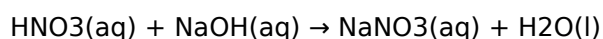
Bases are proton acceptors

Example

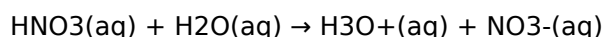


“a” – shows that different ions and molecules are surrounded by solvent molecules, ammonia rather than water.

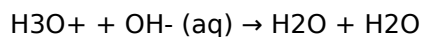
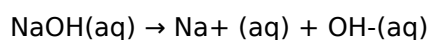
Another reaction is the neutralization of acids with base



Nitric acid solution is a mixture of oxonium ions and nitrate ions;

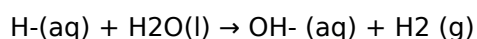


Sodium hydrogen solution is a mixture of sodium ions and hydroxide ions.



Oxonium ion donates a proton to the hydroxide ion

In the reaction of hydride ion and water



H₂O (l) loses proton; so it is bronsted acid

The Lewis theory of acids and bases

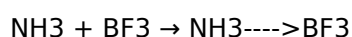
The American chemist decided that more general theory of acids and bases was possible

He stated that;

Acids are electron pair acceptors

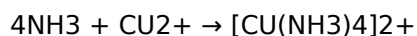
Bases are electron pair donors

This theory was seen to fit with a reaction of ammonia with boron trifluoride in which ammonia function as lewis base



NH₃ donates a lone pair of electron

BF₃ has empty 2p orbital



NH₃ donate lone to the Cu²⁺ ion

[Cu(NH₃)₄]²⁺ Tetraammine copper(ii)

The terms used in describing acids

Strong acids are acids that completely dissociate into ions

Weak acids are acids that partially dissociate into ions

Concentrated acids are acids that have many moles of acid in a litre of solution

Dilute acids are acids that have few moles of acid in a litre of solution

Note;- For acid to be strong, the ion it makes must be energetically stable that is it must not easily change back into molecule from which it came.

The pH of a solution does not necessarily determine the strength of an acid because a solution of high pH may be highly diluted.

$$\text{pH} = -\text{Log}_{10} [\text{H}^+]$$

- Salts

Salts composed of small ions, and typically have high melting and boiling points, and are hard and brittle.

They are usually derivatives of acids

Salts are compounds because they contain at least two elements chemically combined together.

They are ionic compounds i.e. they are made of ions which are atoms or group of atoms that have either a positive or negative charge.

Ions bear a charge due to either loss or gain of one or more of their valence electrons.

Negatively charged ions have gained additional electrons in their outer shell; they are called anions

Positively charged ions have lost electrons; they are called cations

Metals such as sodium, magnesium and calcium have a tendency to donate their electrons which makes them to be cations.

Non-metal like chlorine and most of the halogens readily accept electrons which makes them to be anions

In order to derive any salt chemical formula, the cation and anion must be identified.

For example; table salt NaCl; the salt molecule is comprised of a sodium cation Na^+ and a chloride anion Cl^-

The sodium cation and chloride anion are combined together through an ionic bond.

Salt has a very high solubility in water which is due to high polarity.

Water molecules pull apart the salt cations and anions breaking their ionic bonds

Equation

Properties of salt

- Salt has a crystalline structure and often appears in different colours
- Calcium is white

- Copper sulphate is blue
- Sodium chloride ranges from transparent to white
- The melting and boiling points of salt are extremely high and excess amount of energy is required to break the ionic bonds of salt
- Salts are polar which makes them water soluble and excellent electricity conductors.
- They are good electrolytes because they completely dissociate in water forming ions swimming in water solution.
- Salts taste well and they don't have any odor.
- Salts often form crystal structure or crystal lattice i.e a highly ordered formation of molecules.

Gases

Gas is one of the three classical states of matter

Properties of gases

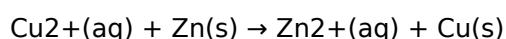
- A gas has no definite shape or volume of its own i.e it acquires the shape of the container and intermolecular attraction is weakest in gases
- A gas has no surface of its own
- A gas is not rigid and is easily compressed
- A gas can diffuse into another gas
- A gas on cooling changes into liquid state
- A gas can flow in all directions
- The molecules are very loosely packed in gases
- Gases are not visible
- Some gases are air; carbon-dioxide, oxygen etc.
- The state of a gas can be completely defined by specifying its temperature, volume, number of moles and pressure.

Redox Reaction

A redox reaction is a reaction in which both oxidation and reduction occur. It involves the transfer of electrons.

The number of electrons lost in oxidation is equal to the number of electrons gained in reduction

Example;



Reduction half reaction;- $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$

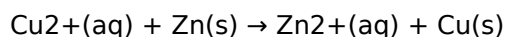
Oxidation half reaction;- $\text{Zn}(\text{s}) - 2\text{e}^- \rightarrow \text{Zn}^{2+}(\text{aq})$

Oxidation is therefore the addition of oxygen and loss of electron

Reduction is therefore the addition of hydrogen and gain of electron

Oxidizing agent [oxidant] is a substance which bring about oxidation; acceptor of electrons and become reduced

Reducing agent [reductant] is a substance which brings about reduction; donor of electrons and become oxidized



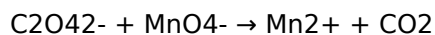
Cu^{2+} ; - oxidizing agent

$\text{Zn}(\text{s})$; - reducing agent

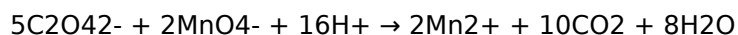
Balancing redox reactions in acidic medium

- H_2O is added to the side that is deficient in oxygen
- Then H^+ is used to make it up

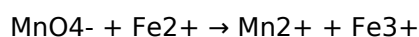
Example 1



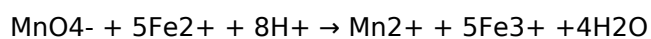
Balanced equation



Example 2



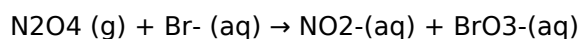
Balanced equation



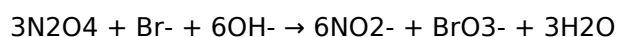
Balancing redox reactions in basic medium

- H_2O is added to the side that is rich in oxygen
- OH^- is used to make it up

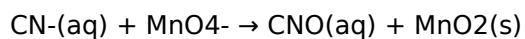
Example 1



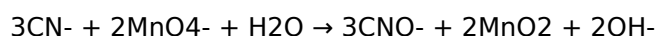
Balanced equation



Example 2



Balanced equation



Thank

you

for

listening

Paper 3 of 7 — CHM 101 material on Electrolysis

Level 100 • First Semester • Handout

CHM 101: GENERAL CHEMISTRY

CREDIT: 3 UNITS

Course Lecturer: ADETORO R.O

1

ELECTROLYSIS

Electrolysis: It is a chemical process where a substance in its molten state or in an aqueous solution is decomposed by the passage of electric current.

Electrolyte: A compound that allows electric current to pass through itself, when either in a molten state or in an aqueous solution, is called an electrolyte. Solutions of sodium chloride, copper sulphate, dilute tetraoxosulphate (vi) acid are examples of strong electrolytes. Weak electrolytes are those compounds which are poor conductors of electricity when they are in a molten state or in an aqueous solution. Electrolytes dissociate to form negatively charged anions and positively charged cations.

-The ions conduct electricity through the electrolyte.

-Cations are attracted towards the negative electrode. They take the excess electrons from the electrode and neutralize themselves. 2

-Anions are attracted towards the positive electrode.

ELECTROLYSIS

Non-electrolyte: A compound which does not allow electric current to pass through itself in any state, molten or aqueous, is called a non-electrolyte.

Electrodes: The strips of metals inserted in the electrolytes for conduction of electricity are called electrodes. The metal electrode connected to the positive terminal of the battery is called the anode (+). The metal electrode connected to the negative terminal of the battery is called the cathode (-).

Electrolytic cell: The complete set-up for electrolysis is called the electrolytic cell. This consists of the vessel containing the electrolyte, anode, cathode, battery and wires. Electrolytic cell is also known as a voltameter, since it generates voltage (or current) at its two terminals.

3

LAWS OF ELECTROLYSIS

Faraday's laws of electrolysis express the dependence on the quantitative or measurable results of electrolysis (i.e. the amount of an element in moles liberated during electrolysis) on the following factors

- i. The time of passing the steady current (in seconds)
- ii. The magnitude of the steady current passed
- iii. The charge on the ion of the element

$$Q = It$$

Q = quantity of electricity, I = current (amps), t = time (seconds)

To investigate factors (i) and (ii), Faraday's first explains these

Faraday's first law of electrolysis states that the mass of a substance liberated at or dissolved from an electrode during electrolysis is directly proportional to the quantity of electricity passing through the electrolyte.

$$M \propto Q$$

$$M = ZQ$$

Where M = mass of the substance liberated, Z = a constant called electrochemical equivalent of the substance, Q =

quantity of electricity

Since $Q = It$

4

LAWS OF ELECTROLYSIS

Then $M = Zit$

To liberate one mole of an element, the least quantity of electricity required is 96500C (1F) and this applies to an element with one +ve charged ions

i.e. $Ag^+ + e^- \rightarrow Ag$

1 F is equivalent to charge of 1 mole of electrons

Faraday's second law of electrolysis states that the number of moles of different ions discharged by the same quantity of electricity is inversely proportional to their respective charges.

e.g. To discharge 1 mole of Cu^{2+} , 2 moles of electrons are required



Combining the two laws, the following relationship is established

Number of moles of products formed = $\frac{It}{nF}$

nF

Where F is Faraday's constant, n = number of moles of electrons required to form 1 mole of products

5

EXAMPLES

Examples

1. How many Faraday's passes through a resistance in a circuit carrying a current of 5A for 1 hr (1F = 96500C)

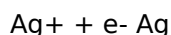
$$Q = It$$

$$= 5 \times 60 \times 60 = 18000C$$

$$\text{No of Faradays} = \frac{18000}{96500} = 0.187F$$

96500

2. What is the number of moles of silver nitrate formed when a current of 0.3A is passed through a silver nitrate solution for 30 min



No of moles of Ag formed = $\frac{It}{nF}$

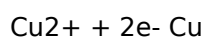
No of moles of Ag formed is $\frac{0.3 \times 30 \times 60}{1 \times 96500}$

= 5.6×10^{-3} mol

6

EXAMPLES

1. What is the mass of copper formed at the cathode when a current of 0.25A is passed through a copper(II)sulphate solution for 1 hr (molar mass Cu = 63.5)



No of moles of Cu formed is $\frac{0.25 \times 60 \times 60}{2 \times 96500}$

= 4.66×10^{-3} mol

Mass of Cu formed = $4.66 \times 10^{-3} \times 63.5$

= 0.296g

TUTORIAL QUESTIONS

1. What is the time required to pass 1 Faraday of electricity through an electrolyte with a current of 0.35A

2. What current in amperes is required in order to deposit 6.35g of copper in 50 min from a copper(II)sulphate solution

3. Calculate the mass of aluminium that would be deposited during the electrolysis of a molten aluminium salt by a current of 10A for 5 hr

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Paper 4 of 7 — CHM 101 introductory chemistry I new

Level 100 • First Semester • Handout

CHM 101

INTRODUCTORY CHEMISTRY I

COURSE LECTURER: Azeez Luqmon A., PhD.,

1

COURSE SYNOPSIS

❖ Hybridization and shapes of molecules

❖ Shapes of solids

2

Course Objectives

With this course, students will

> Understand comprehensively concepts of hybridization and be able to predict and draw shapes of molecules

> Understand shapes of solids

3

Course Schedule

> Week 1 : Hybridization, shapes of molecules

> Week 2: Shapes of ionic solids

4

HYBRIDIZATION AND SHAPES OF MOLECULES

Valence Shell Electron Pair Repulsion Model (VSEPR): uses electron pair of valence shells to predict shapes

The total number of valence shell electron pairs decides the shape of the molecule.

The electron pairs have a tendency to orient themselves in a way that minimizes the electron-electron repulsion between them and maximizes the distance between them

Number of Electron pair Bonding pairs Lone pairs Shape Angle

2 2 0 Linear 180 o

3 3 0 Trigonal planar 120 o

4 4 0 Tetrahedral 109.5 o

4 3 1 Trigonal pyramidal 106.7 o

4 2 2 Angular 104.5 o

5 5 0 Trigonal bipyramidal Axial 90 o ,Equatorial 120o

0 6 0 Octahedral 90 o 5

VSEPR Rules

Identify the central atom

Write its electronic

configuration to get valence
electrons Add one electron for each
bonding atom
Add or subtract electrons for
charge
Add up all electrons and
calculate the electron pairs
Compare the number of
electron pairs with the
shapes in the table SAMPLE HEADLINE
This is a sample text that
you can edit.

VSEPR Rules

BCl₃

B- central atom - 2,3

Valence electrons - 3

Number of bonding atoms - 3

Total - 6

Electron pair = $6/2 = 3$ - trigonal planar

CH₄

C- central atom - 2,4

Valence electrons - 4

Number of bonding atoms - 4

Total - 8

Electron pair = $8/2 = 4$ - tetrahedral.

NH₃

H₂O -

N- central atom - 2,5

O- central atom - 2,6

Valence electrons - 5

Valence electrons - 6

Number of bonding atoms - 3

Number of bonding atoms - 2

Total - 8

Total - 8

Electron pair = $8/2 = 4$ - tetrahedral. But only three

Electron pair = $8/2 = 4$ - tetrahedral. But only three

atoms are bonded, thus, it has a lone pair of

atoms are bonded, thus, it has 2 lone pairs of

electron. The real shape is trigonal pyramidal

electron. The real shape is angular

8

PCl₅

C

I

P- central atom - 2,8,5

C

I

Valence electrons - 5

C

I P

Number of bonding atoms - 5

Total - 10

C

I

Electron pair = $10/2 = 5$ - trigonal bipyramidal. C

I

SF₆ F

S- central atom - 2,8,6

F

Valence electrons - 6 F

S

Number of bonding atoms - 6

Total - 12

F F

Electron pair = $12/2 = 6$ - Octahedral F

9

Valence Bond Theory: it uses hybridization theory to predict shapes

Hybridization - the concept of mixing atomic orbitals to form new hybrid orbitals suitable for the qualitative description of atomic bonding properties

Hybridized orbital Bonding/lone pairs Shape

sp 2/0 Linear

sp² 3/0 Trigonal planarsp³ 4/0 Tetrahedralsp³ 3/1 Trigonal pyramidalsp³ 2/2 Angularsp³d 5/0 Trigonal bipyramidalsp³d² 6/0 Octahedral

10

BF₃ - 1s² 2s² 2p¹

11

CH₄ – 1s² 2s² 2p²

12

NH₃ – 1s² 2s² 2p³

13

H₂O – 1s² 2s² 2p⁴

14

CO₂ – 1s² 2s² 2p²

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SHAPES OF IONIC SOLIDS

The structure of ionic solids is determined by how the cations and anions can pack together. Generally, one of the ions adopts a standard packing structure, like the metal atoms in a metallic solid. The counterions then fit into the holes or interstitial spaces among these ions.

Ionic crystals consist of a three-dimensional arrangement of ions held in a fixed positions by strong electrostatic forces

Common type:

- > Face-centered cubic (FCC)
- > Body-centered cubic (BCC)
- > Hexagonal closed-packed (HCP) 16

Crystal structures

Metallic crystal structures

Face-Centered Cubic (FCC) structure has a cubic geometry with atoms located at each of the corners and

the centers of all the cube faces.

Many metals have this FCC structure.

Callister, 2005 17

The relation between the cube edge length, a and the atomic radius, R :

$$a = 2R\sqrt{2}$$

1 1

Total atom per unit cell of FCC is 4, i.e. = $8 \left(\frac{1}{8}\right) + (6 \left(\frac{1}{2}\right))$

8 2

The coordination number (CN) is a number that an atom touches the nearest neighbour atoms.

The coordination number of FCC is 12

Body-Centered Cubic structure has a cubic unit cell with atoms located at each of the corners and a single

atom at the cube center.

18

The relation between the cube edge length, a and the atomic radius, R :

$$4R$$

a=

3

1

Total atom per unit cell of BCC is 2, i.e. = $8 \times \frac{1}{8} + (1 \times 1)$

8

The coordination number of BCC is 8

Hexagonal Closed-Pack structure is the common metallic crystal structure.

It has 6 atoms in regular hexagones at the top and the bottom faces and surround a single atom in the center. Three additional atoms are on a plane located between the top and the bottom planes.

19

1 1

Total atom per unit cell of HCP is 6, i.e. = $2 \times 6 \times \frac{1}{6} + 2 \times \frac{1}{2} + (1 \times 3)$

6 2

The coordination number of HCP is 12

20

Paper 5 of 7 — CHM101 Lecture Note on Thermo Chemical kinetics

Level 100 • First Semester • Handout

CHM 101: Thermochemistry & Chemical Kinetics Lecture Note

CHM 101: THERMOCHEMISRTY & CHEMICAL KINETICS

COURSE UNIT: 3

COURSE SYNOPSIS

Thermochemistry: Standard enthalpy changes of reaction, formation, combustion and neutralization. Hess law, lattice energy for simple ionic crystals.

Chemical Kinetics: Simple equations; order of reactions; rate constants; simple calculations on half-life. Qualitative effects of temperature on rate constant; catalysis.

Course Lecturer: Dr. A. J. Anifowose

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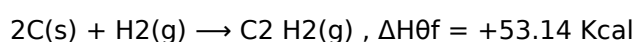
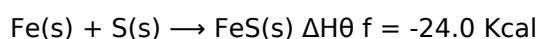
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What is Thermochemistry?

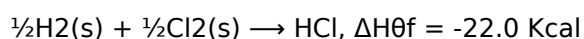
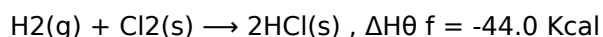
Thermochemistry is an aspect of chemistry that deals with the amount of heat exchange (i.e., absorbed or released) during a chemical reaction

Types of Heat (Enthalpy) of Reaction

Heat of formation is defined as the change in enthalpy that takes place when one mole of the compound is formed from its element. It is denoted by change in H_f if the heat of formation of ferrous sulphide and acetylene may be expressed as



Similarly, the reaction between gaseous hydrogen and gaseous chlorine to form gaseous hydrogen chloride is represented by the equation:



Standard Heat of Formation: The standard heat of formation of a compound is defined as the change in enthalpy that takes place when one mole of the compound is formed from its element, all substances being in their standard state (298k and 1atm pressure).

Heat of reaction or Enthalpy of reaction: Heat of reaction may be defined as the amount of heat absorbed or evolved in a reaction when the number of moles of reactants, as represented by the balanced chemical equations, changed completely into

the product.



The heat exchange accompanying a reaction taking place at 298k and one atmospheric pressure is called the Standard Heat change or Standard Enthalpy change. It's denoted by change in ΔH^θ

Standard Heat of Reaction (ΔH^θ) from standard heat of formation (ΔH^θ_f)

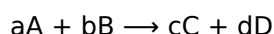
The standard heat of reaction is equal to the standard heat of formation of product—the standard heat of formation of reactant

$$\Delta H^\theta = \Delta H^\theta_f(\text{products}) \rightarrow \Delta H^\theta_f(\text{reactants})$$

2

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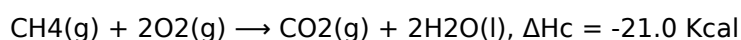
Let us consider the general reaction



$$\Delta H^\theta = \Delta H^\theta_f(\text{products}) \rightarrow \Delta H^\theta_f(\text{reactants})$$

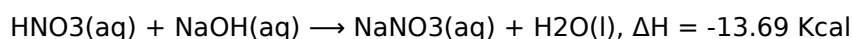
$$= [c \cdot \Delta H^\theta_f(C) + d \cdot \Delta H^\theta_f(D)] \rightarrow [a \cdot \Delta H^\theta_f(A) + b \cdot \Delta H^\theta_f(B)]$$

Heat of Combustion: The heat of combustion of a substance is defined as the change in enthalpy of a system when one mole of the substance is completely burnt in excess of air or oxygen. It is denoted by change in H_C



Heat of combustion is always -ve

Heat of Neutralization: Heat of neutralization is defined as the change in heat content (enthalpy) of the system when one form equivalent of an acid is neutralized by one form equivalent of a base or vice versa in dilute solution

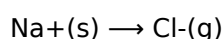
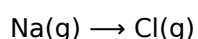
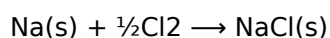


HESS LAW

Hess law state that if a chemical change can be made to take place in two or more different ways whether in one step or two or more steps, the amount of total heat changes is same, no matter by which method the change is brought about.

LATTICE ENERGY

The positive and negative ion in an ionic crystal are held together by electrostatic forces. The bond energy is expressed in terms of the lattice energy which may be defined as the change in enthalpy (heat change) that occurs when one mole of a solid crystalline substance is formed from its gaseous ions.



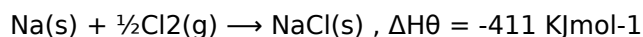
A Born-Haber cycle for the formation of NaCl crystal from its element

3

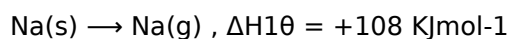
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Determination of Lattice Energy

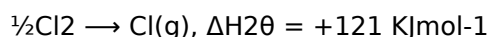
The lattice energy of an ionic crystal can be found by applying Hess law. Consider an enthalpy change for the direct formation of NaCl



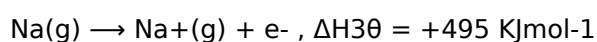
Step 1: conversion of sodium metal to gaseous atom



Step 2: Dissociation of chlorine molecules to chlorine atoms



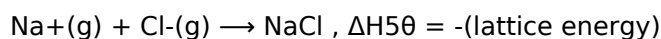
Step 3: Enthalpy of ionization; conversion of gaseous sodium to sodium ion by loss of an electron



Step 4: Chlorine atom gains an electron to form chlorine ion. The energy of electron affinity of chlorine



Step 5: Sodium and chloride ions get together and form the crystal lattice



$$\Delta H_{10} + \Delta H_{20} + \Delta H_{30} + \Delta H_{40} + \Delta H_{50} = -411 \text{ kJ}$$

$$108 + 121 + 495 - 348 - \text{lattice energy} = -411$$

$$\text{Lattice energy} = +787 \text{ KJmol}^{-1}.$$

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CHEMICAL KINETICS

This is the study of the rate of reaction in chemistry and it covers the following.

1. The rate of reactions and rate laws;
2. The factors as temperature, pressure, concentration and catalyst that influences the rate of reaction;
3. The mechanism or the sequence of steps by which a reaction occurs.

The rate of chemical reaction depends on the nature of reactant, temperature, concentration or pressure of the reactant, surface area of the reactants and the presence of catalyst

1. Nature of the reactants: The complexity of the reactant will affect their ability to collide rightly.
2. Effect of temperature: A chemical reaction is affected by temperature because the velocity of the reacting molecules is altered by change in temperature. Increase in temperature increase the rate of reacting molecule. According to Maxwell-Boltzman's molecular velocity distribution, higher temperature activates more molecules to attain high velocity which is required for effective collision
3. Effect of pressure/concentration: With the exception of zero order reaction, the rate

of chemical reaction is directly proportional to the concentration of the reacting specie or to the pressure of the reacting species (if the reactants are gaseous).

4. Effect of Surface Area: This is important only if the system is heterogenous. The larger the surface area for contact, the more molecules that are exposed to each other and the more the number of possible collisions.

5. Effect of catalyst: The presence of catalyst can increase or decrease the speed of a reaction. For instance, hydrogen and oxygen do not combine but in the presence of platinum catalyst. Similarly, a small amount of glycerin slows down the decomposition of hydrogen peroxide.

In chemical kinetics, reaction can be categorized into two:

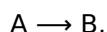
1. Homogenous reaction: It takes places entirely in one phase.
2. Heterogeneous: It takes place in two or more phases. For example, gaseous reaction taking place on the surface of a solid catalyst.

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RATE OF REACTION

The rate of a reaction tells us the speed of reaction. Consider a simple reaction:



The concentration of reactant A decreases and that of B increases as time passes.

The rate of reaction is defined as the change in concentration of reactant or product per unit time. For the given reaction. The rate of reaction may be equal to the rate of disappearance of A which is equal to the rate of appearance of B. Therefore,
 $\text{rate} = -d[A]/dt = +d[B]/dt.$

Where square bracket represents concentration of A and B. The negative sign shows the concentration of A decreases in the course of the reaction. Whereas the positive sign indicates increase in the concentration of B as the reaction proceeds.

THE RATE LAW

At a fixed temperature, the rate of a given reaction depends on concentration of reactant. The exact relationship between concentration and rate is determined by measuring the reaction rate with different initial reactant concentration. It is shown that the rate of reaction is directly proportional to the reaction concentration, each concentration being raised to some powers.

Thus, for a substance A undergoing reaction: rate, $r \propto [A]^n$, $r = k[A]^n$

For a reaction: $2A + B \longrightarrow \text{product}$,

the reaction rate with respect to A or B is determined by varying the concentration of one reactant, keeping that of the other. Thus, the rate of reaction can be expressed as
 $\text{rate} = k[A]^m[B]^n$

An expression which shows how the rate of reaction is related to concentration of reactant is called the rate law or rate equation.

The power (exponent) of concentration n or m in the rate law is usually small whole number integers 1,2,3 or fractional. The proportionality constant, k, is called the rate

constant for the reaction

ORDER OF REACTION

It is defined as the sum of the powers of the concentration in the rate law. Consider the example of a reaction which has the rate law: $\text{rate} = k[A]^m[B]^n$.

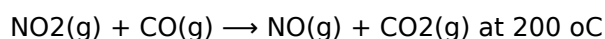
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The order of such a reaction is $m + n$.

The order of a reaction can also be defined with respect to a single reactant. Thus, the reaction order with respect to A is m , and n with respect to reactant B. the overall order of the reaction ($m + n$) may range from 1 to 3 and can be fractional. Examples of order of reaction.

The reaction will be classified as first order ($m + n = 1$); second order ($m + n = 2$); third order, if $m + n = 3$. A zero order reaction is the reaction which does not depend on the concentration of the reactant. A reactant whose concentration does not affect the reaction rate is not included in a rate law.



$$\text{rate} = k[\text{NO}_2]^2$$

Here, the rate does not depend on the concentration of CO, so it (CO) is not included in the rate law and the power of CO is understood to be zero. Therefore, the reaction is zero order with respect to CO, but second order with respect to NO₂. The overall reaction order is $2+0 = 2$.

MOLECULARITY OF A REACTION

Molecularity of an elementary reaction is defined as the number of reactant molecules involved in a reaction.

Differences between order and molecularity of a reaction

S/N Order or a reaction Molecularity of a reaction

1 It is the sum of powers of the concentration terms in the rate law It is no of reacting species undergoing simultaneously collision in the expression elementary or simple reaction

2 It is an experimentally determined value It is a theoretical concept

3 It can have fractional value It is always a whole number

4 It can assume zero value It cannot have zero value

5 Order of a reaction can change with reaction condition for a the reaction condition such as given chemical equation pressure, temperature concentration

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METHODS FOR THE DETERMINATION OF THE ORDER OF REACTION AND ITS RATE CONSTANT

(a) The Inspection method

Examples:

(i) For a reaction represented by the equation

$A(g) + B(g) \rightarrow C(g)$, the following data were obtained

Experiment [A] (M) [B⁻] (M) rate (Ms⁻¹)

1 1.0 1.0 2.0

2 1.0 2.0 4.0

3 2.0 1.0 4.0

The data show that three experiments were carried out such that the concentrations of A and B are kept constant in two of the experiments. Looking at the data when [A] was maintained constant and [B] was doubled (experiment 1 and 2), the rate of reaction doubled. This implies that the rate is directly proportional to [B]. Hence, the order with respect to B is 1. Similarly, when [B] was kept constant and [A] was doubled (experiments 1 and 3), the rate of reaction doubled. Hence, rate was also first order in [A]. Therefore, the overall order of the reaction was 2. The rate constant, k, can be calculated by substituting a pair of values in the rate equation: $\text{rate} = k[A][B]$.

NOTE: If you double a reactant concentration for a second-order, the rate should increase four-fold and for a third-order, the rate should increase eight-fold.

(ii) For a reaction: $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

the initial rates of the reaction were determined for different concentrations of $N_2O_5(g)$

[N₂O₅] (mol/L) Rate (mol/L/h)

0.1 0.016

0.2 0.032

0.4 0.064

The data indicate that the rate is doubled as the concentration is doubled. Thus, $\text{rate} \propto [N_2O_5]^1$.

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The reaction is therefore first-order and the rate equation is $\text{rate} = k[N_2O_5]^1$. The rate constant for the reaction is obtained by substitution of any pair of values from the table into the rate reaction:

$$k = \text{rate}/[N_2O_5]$$

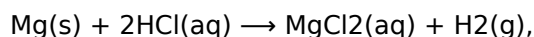
$$k = 0.016/0.1, k = 0.16 \text{ h}^{-1}.$$

(b) The rate curve method

The rate at a given concentration is the slope of the tangent to the concentration-time curve at that concentration. By assuming an order of reaction, a plot of the rate against concentration raised to the power which is equal to the assumed order is carried out. If the assumed order tallies with the actual order of the reaction then a linear plot will be

obtained.

For a reaction:



The rate curve is obtained by plotting $[\text{HCl}]$ (mol/L) against time. The reaction rate obtained from rate curve is then plotted against $[\text{HCl}]^n$, where n stands for the assumed order. If a linear plot is obtained, then n represents the actual order of the reaction.

Experimentally, the value of n was found to be 2. The rate constant for the reaction can be obtained by substituting a pair of values into the rate equation:

$$\text{Rate} = k[\text{HCl}]^2.$$

(c) Integrated rates method (for your further read)

RATE OF REACTION

Introduction: The rate of a reaction is the rate at which products are formed or the rate at which reactants are used up in the reaction. Rate of reaction is the change in the concentration of a reactant or product with time. By definition, it is necessary to monitor the concentration of the reactant (or the product) as a function of time.

Consider a reaction of molecular bromine and formic acid:



$$\text{Average rate} = \Delta[\text{Br}_2]/\Delta T$$

$$= -([\text{Br}_2]_{\text{final}} - [\text{Br}_2]_{\text{initial}})/(\text{T}_{\text{final}} - \text{T}_{\text{initial}})$$

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Molecular Bromine is reddish brown in color. All other species in the reaction are colourless. As the reaction progresses, the concentration of Bromine steadily decreases and its colour fades. This loss of colour and, hence, concentration can be monitored easily with a spectrometer, which registers the amount of visible light absorbed by bromine. Measuring the change (decreases) in bromine concentration at some initial time and then at some final time enables us to determine the average rate of the reaction during that interval.

The rate of any reaction depends on a number of factors, among which are concentration or pressure of reactants, temperature, presence of catalyst and nature of reactants. For instance, rate of reaction increases as the surface of the reactant is increased and this is why solid reactant reacts with less rate than powdered reactant.

Question

Using the data provided below

- Calculate the average rate over the first 50secs.
- Calculate the average rate over the first 100secs.

Time (s) $[\text{Br}_2]$ (M) Rate(M/s) $k = \text{rate(s}^{-1})/[\text{Br}_2]$

0 0.012 4.2×10^{-5} 3.50×10^{-3}

50 0.0101 3.52×10^{-5} 3.49×10^{-3}

100 0.00846 2.96×10^{-5} 3.50×10^{-3}

$$150 \quad 0.00710 \quad 2.49 \times 10^{-5} \quad 3.51 \times 10^{-3}$$

$$250 \quad 0.00500 \quad 1.75 \times 10^{-5} \quad 3.50 \times 10^{-3}$$

(i) average rate over the first 50sec:

$$\text{rate} = -(0.0101 - 0.012)/(50 - 0)$$

$$\text{rate} = 1.9 \times 10^{-3}/50 = 3.8 \times 10^{-5} \text{ M/s}$$

(ii) calculate on your own following worked example in (i) above

Let us consider the effect the bromine concentration has on the rate of reaction. Looking at the data, the concentration at $t = 50 \text{ s}$ is double the concentration at $t = 250 \text{ s}$ and the rate of reaction at $t = 50 \text{ sec}$ is double the rate at $t = 250 \text{ sec}$. Thus, as the concentration of bromine is doubled the rate of the reaction also doubles therefore, the rate of reaction is directly proportional to the bromine concentration

$$\text{rate} \propto [\text{Br}_2], \text{ thus } r = k [\text{Br}_2]$$

Where k is known as the rate constant

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Because the rate of reaction has the unit of M/s and bromine concentration is in M , the unit of k is " s^{-1} ". It is important to note that k is not affected by bromine concentration. To be sure, the rate is greater at a higher concentration and smaller at lower concentration of bromine, but the ratio of $\text{rate}/[\text{Br}_2]$ remains the same provided the temperature does not change.

Reaction rate and Stoichiometry

We have seen that for stoichiometrically simple reactions of type $\text{A} \rightarrow \text{B}$, the rate can either be expressed in terms of decrease in the reactant concentration or increase in product concentration. For more complex reactions, we must be careful in writing the rate expression.

Consider for example, $2\text{A} \rightarrow \text{B}$:



$$\text{Rate} = -\frac{1}{2} \frac{\Delta[\text{A}]}{\Delta t} = \frac{\Delta[\text{B}]}{\Delta t}$$

$$\text{Rate} = \frac{\Delta[\text{B}]}{\Delta t}$$

$$\text{Rate} = \frac{\Delta[\text{B}]}{\Delta t}$$

$$r = \frac{\Delta[\text{B}]}{\Delta t}$$

$$r = \frac{\Delta[\text{B}]}{\Delta t}$$

In general, for the reaction $a\text{A} + b\text{B} \rightarrow c\text{C} + d\text{D}$. The rate is given by:

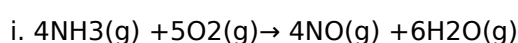
$$-\frac{1}{a} \frac{\Delta[\text{A}]}{\Delta t} = -\frac{1}{b} \frac{\Delta[\text{B}]}{\Delta t} = \frac{1}{c} \frac{\Delta[\text{C}]}{\Delta t} = \frac{1}{d} \frac{\Delta[\text{D}]}{\Delta t}$$

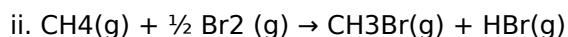
$$= \frac{\Delta[\text{B}]}{\Delta t} = \frac{\Delta[\text{C}]}{\Delta t} = \frac{\Delta[\text{D}]}{\Delta t}$$

$$\frac{\Delta[\text{B}]}{\Delta t} = \frac{\Delta[\text{C}]}{\Delta t} = \frac{\Delta[\text{D}]}{\Delta t}$$

Question

Find the rate expression for the following reaction in terms of disappearance and appearance of the product.





Answer:

$$\Delta[\text{CH}_4] = \Delta[\text{CH}_3\text{Br}] = \Delta[\text{HBr}] = -\Delta[\text{Br}_2]$$

i. $\Delta[\text{CH}_4] = \Delta[\text{CH}_3\text{Br}] = \Delta[\text{HBr}] = -\Delta[\text{Br}_2]$

$$\Delta[\text{CH}_4] = \Delta[\text{CH}_3\text{Br}] = \Delta[\text{HBr}] = -\Delta[\text{Br}_2]$$

ii. calculate on your own following the solved question in (i) above

Question 1. Consider the reaction: $4\text{NO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{N}_2\text{O}_5(\text{g})$

Suppose that at a particular moment during the reaction molecular oxygen is reacting as the rate of 0.0024 Ms^{-1}

a) At what rate is N_2O_5 being formed?

b) At what rate is NO_2 reacting?

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Answer

a. $r = \frac{1}{2} \Delta [\text{N}_2\text{O}_5] / \Delta t$

$$0.0024 = \frac{1}{2} \Delta [\text{N}_2\text{O}_5] / \Delta t$$

$$\Delta [\text{N}_2\text{O}_5] / \Delta t = 0.0024 \times 2$$

$$\Delta [\text{N}_2\text{O}_5] / \Delta t = 0.0048 \text{ M/s}$$

b. $r = -1/4 \Delta [\text{NO}_2] / \Delta t$

$$0.0024 = -1/4 \Delta [\text{NO}_2] / \Delta t$$

$$\Delta [\text{NO}_2] / \Delta t = 4 \times 0.0024$$

$$\Delta [\text{NO}_2] / \Delta t = -9.6 \times 10^{-3} \text{ M/s}$$

RATE LAW OF FIRST ORDER REACTION

$\text{A} \rightarrow \text{Product}$

Suppose at the beginning of a reaction time ($t = 0$), the concentration of $\text{A} = a \text{ mol/L}$. If after time t , $x \text{ mol}$ of A has changed, the concentration of A is $a-x$.

We know that for a first order reaction, the rate of reaction (dx/dt) is directly proportional to the concentration of the reactant. Thus,

$$dx/dt = k(a-x)$$

$$dx/(a-x) = k dt \dots \dots \dots (1)$$

$$\int \frac{dx}{a-x} = \int k dt$$

Integration of equation (1) gives $\ln(a-x) = -kt + \ln a$

$$(\ln(a-x) - \ln a) = -kt$$

$$-\ln(a-x) + \ln a = kt \dots \dots \dots (ii)$$

$$k = 2.303/t \log (a/a-x)$$

$$k = 1/t \ln (a/a-x)$$

In terms of volume,

$$k = 2.303/t \log (V_0/V_0-V_t)$$

Calculation of half-life of first order reaction

$$k = (2.303/t) \log ([A]_0/[A])$$

Where $[A]_0$ is the initial concentration of A; $[A]$ is the concentration of A with respect to time.

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Half life, $t_{1/2}$ is the time when initial concentration of A reduces to half.

A reduces to half i.e $[A] = \frac{1}{2}[A]_0$, substituting value in the integration rate equation we have,

$$k = 2.303/t \log ([A]_0/\frac{1}{2}[A]_0)$$

$$k = 2.303/t_{1/2} \log 2$$

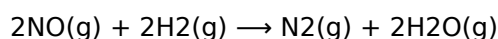
$$t_{1/2} = 2.303 \log 2/k$$

$$t_{1/2} = (2.303 \times 0.3010)/k$$

$$t_{1/2} = 0.693/k$$

Rate law and order of reaction

Q1: The reaction of nitric oxide (NO) with hydrogen at 1280°C



From the following data collected at this temperature, determine (i) the rate law (ii) the rate constant (iii) the rate of the reaction when the concentration of NO = 12×10^{-3} M and concentration of H₂ = 6×10^{-3} M

Experiment [NO] (M) [H₂] (M) Initial rate (M/s)

1 5×10^{-3} 2×10^{-3} 1.3×10^{-5}

2 10×10^{-3} 2×10^{-3} 5×10^{-5}

3 10×10^{-3} 4×10^{-3} 10×10^{-5}

Answer:

Generally, the rate law for the reaction is written as: $r = k[\text{NO}]^x[\text{H}_2]^y$

In Exp 1, $r = 1.3 \times 10^{-5}$, then

$$1.3 \times 10^{-5} = k(5 \times 10^{-3})^x(2 \times 10^{-3})^y \dots\dots\dots(1)$$

In Exp 2, $r = 5 \times 10^{-5}$

$$5 \times 10^{-5} = k(10 \times 10^{-3})^x(2 \times 10^{-3})^y \dots\dots\dots(2)$$

In Exp 3, $r = 10 \times 10^{-5}$

$$10 \times 10^{-5} = k(10 \times 10^{-3})^x(4 \times 10^{-3})^y \dots\dots\dots(2)$$

Equation 2 ÷ 1 results:

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$$5 \times 10^{-5} \div 1.3 \times 10^{-5} = (10 \times 10^{-3})^x \div (5 \times 10^{-3})^x \times (2 \times 10^{-3})^y \div (2 \times 10^{-3})^y$$

$$1.3 \times 10^{-5} \div 5 \times 10^{-5} = (10 \times 10^{-3})^x \div (5 \times 10^{-3})^x \times (2 \times 10^{-3})^y \div (2 \times 10^{-3})^y$$

$$10^{-1} = 2^y$$

$$3.85 = 2^y$$

$$5 = 2^y$$

$$2^y = 2^2$$

$$y = 2$$

Equation 3÷2 results:

$$10 \times 10^{-5} \div (10 \times 10^{-5}) \div (4 \times 10^{-3})$$

$$5 \times 10^{-5} \div (10 \times 10^{-5}) \div (2 \times 10^{-3})$$

$$2 = 2$$

$$= 1$$

Hence, the rate law for the reaction is, $r = k[\text{NO}]^2[\text{H}_2]$

(ii) from equation 1,

$$1.3 \times 10^{-5} = k(5 \times 10^{-3})^2(2 \times 10^{-3})$$

$$1.3 \times 10^{-5} = k(25 \times 10^{-6})(2 \times 10^{-3})$$

$$(25 \times 10^{-6}) \times (2 \times 10^{-3})$$

$$k = 260 \text{ M}^{-2} \text{ s}^{-1}.$$

(iii) from the rate law, $r = k[\text{NO}]^2[\text{H}_2]$,

When $[\text{NO}] = 12 \times 10^{-3} \text{ M}$ and $[\text{H}_2] = 6 \times 10^{-3} \text{ M}$

Then,

$$r = 260 (12 \times 10^{-3})^2(6 \times 10^{-3})$$

$$r = 2.25 \times 10^{-4} \text{ M s}^{-1}.$$

Q2: The reaction of $\text{S}_2\text{O}_8^{2-}$ with iodide is $\text{S}_2\text{O}_8^{2-}(\text{aq}) + 3\text{I}^-(\text{aq}) \rightarrow 2\text{SO}_4^{2-}(\text{aq}) + \text{I}_3^-(\text{aq})$

From the following data collected at a certain temperature determine the rate law

i. the rate constant

ii. the order of reaction

iii. the rate of reaction when the initial concentration of $\text{S}_2\text{O}_8^{2-}$ and I^- are halved and doubled respectively

iv. Compare the rate of reaction I question (4) obtained in experiment (1)

Experiment $[\text{S}_2\text{O}_8^{2-}] \text{ (M)}$ $[\text{I}^-] \text{ (M)}$ Initial rate (M/s)

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1 0.08 0.034 2.2×10^{-4}

2 0.08 0.017 1.1×10^{-4}

3 0.16 0.017 2.2×10^{-4}

Summary of Kinetics of Zero order, first order and second order reactions

Order Rate law Conc.-time Half-life ($t_{1/2}$)

equation

$$0 \text{ rate} = k [\text{A}] t = -kt + [\text{A}]_0 \quad t_{1/2} = [\text{A}]_0 / 2k$$

$$1 \text{ rate} = k[\text{A}] \ln([\text{A}]_t / [\text{A}]_0) = -kt \quad t_{1/2} = 0.693/k$$

$$2 \text{ rate} = k[\text{A}]^2 \quad (1/[\text{A}])_t = kt + (1/[\text{A}])_0 \quad t_{1/2} = 1/k[\text{A}]_0$$

$$(1/[\text{A}])_0$$

Q3: The conversion of cyclopropane to propene in the gas phase is a first order reaction with a rate constant of $6.7 \times 10^{-4} \text{ s}^{-1}$ at 500°C

a) If the initial concentration of cyclopropane was 0.25 M , what is the concentration after 8.8 min ?

b) How long (in minutes) will it take the concentration of cyclopropane to decrease from 0.25 to 0.15 M

c) How long (in mins) will it take to convert 74% of the starting material

Answer:

(a) Since the reaction is a first order and concentration-time dependent, the equation $\ln([A]_t/[A]_o) = -kt$ applies to solve the problem

$$[A]_t = ?$$

$$[A]_o = 0.25$$

$$k = 6.7 \times 10^{-4} \text{ s}^{-1}$$

$$t = 8.8 \text{ min (} = 528 \text{ s)}$$

$$\ln([A]_t/0.25) = -(6.7 \times 10^{-4} \times 528)$$

$$([A]_t/0.25) = e^{-0.3538}$$

$$= 0.176 \text{ M}$$

(b) $\ln([A]_t/[A]_o) = -kt$ which is the same as $\ln[A]_t - \ln[A]_o = -kt$

$$\ln 0.15 - \ln 0.25 = -(6.7 \times 10^{-4})t$$

$$-0.5108 = -(6.7 \times 10^{-4})t$$

$$t = 762.4 \text{ s (or 12.7 min)}$$

(c) If at time t, 74% of the starting material is converted, then what remains as $[A]_t$ is 26% of 0.25 M

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Hence, $[A]_t = 0.065 \text{ M}$

$$\ln[A]_t - \ln[A]_o = -kt$$

$$\ln 0.065 - \ln 0.25 = -(6.7 \times 10^{-4})t$$

$$-1.347 = -(6.7 \times 10^{-4})t$$

$$t = 33.5 \text{ min.}$$

THE EFFECT OF TEMPERATURE ON THE RATE OF REACTION

The rate of reaction doubles for every 10K rise in temperature. The relationship between temperature and rate constant is given by Arrhenius equation which is expressed as:

$k = Ae^{-E_a/RT}$, where K is the rate constant for the reaction, A is the Arrhenius constant, E_a is the activation energy, R is the gas constant and T is the temperature in Kelvins.

$$\log k = \log A - E_a/(2.303RT)$$

A graph of $\log K$ against $1/T$ is linear and E_a and A can be calculated from the slope and intercept of the graph, respectively.

By determining the rate constant at two different temperatures, you obtain:

$$\log(k_2/k_1) = (E_a/2.303R)(1/T_1 - 1/T_2).$$

E_a can be calculated from the expression.

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Further Reading

- Carey F. A. and Giuliano R. M. (2014) Organic Chemistry, 9th edn. McGraw-Hill Education, New York.
- Bahl A, Bahl B.S and Tull G.D (2012). Essentials of Physical Chemistry, S. Chand & Company LTD, 2012 Edition, New- Delhi.

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Paper 6 of 7 — CHM 101 LECTURE 1

Level 100 • First Semester • Handout

CHM 101 COURSE SYNOPSIS

Atoms, molecules, elements and compounds and chemical reactions. Modern electronic theory of atoms. Electronic configuration, periodicity and building up of the periodic table. Hybridization and shapes of simple molecules. Valence Forces. Structure of solids. Chemical equations and stoichiometry;

Chemical bonding and intermolecular forces, kinetic theory of matter. Elementary thermochemistry.

Rates of reaction, equilibrium and thermodynamics. Acids, bases and salts. Properties of gases. Redox

reactions and introduction to electrochemistry. Radioactivity.

ATOM

An atom is the defining structure of an element, which cannot be broken by any chemical means.

A typical atom consists of a nucleus of positively-charged protons and electrically neutral neutrons with

negatively-charged electrons orbiting this nucleus. However, an atom can consist of a single proton (i.e.,

the protium isotope of hydrogen) as a nucleus. The number of protons defines the identity of an atom or

its element. Examples of atoms are hydrogen, Carbon.

Bohr model and

Electron cloud model

Atom Size, Mass, and Charge

The size of an atom depends on how many protons and neutrons it has, as well as whether or not it has electrons. A typical atom size is around 100 picometers or about one ten-billionth of a meter. Most of the volume is empty space, with regions in which electrons may be found. Small atoms tend to be spherically symmetrical, but this is not always true of larger atoms. Contrary to most diagrams of atoms, electrons do not always orbit the nucleus in circles.

Atoms can range in mass from 1.67×10^{-27} kg (for hydrogen) to 4.52×10^{-25} kg for superheavy radioactive nuclei. The mass is almost entirely due to protons and neutrons, as electrons contribute negligible mass to an atom.

An atom that has an equal number of protons and electrons has no net electrical charge. An imbalance in the numbers of protons and electrons forms an atomic ion. So, atoms may be neutral, positive, or negative.

MOLECULES

A molecule consists of more than one atom. A given

molecule has a constant composition, which means the ratio of different atoms is always the same. Identical molecules also display the same chemical properties. Unlike atoms, molecules may be broken by chemical reactions. Breaking all of the chemical bonds in a molecule results in a collection of atoms

Examples of Molecules

A molecule may consist of two or more atoms of a single element or atoms of different elements. Here are some examples of molecules:

- H₂O (water)
- N₂ (nitrogen)
- O₃ (ozone)
- CaO (calcium oxide)
- CO₂ (carbon dioxide)
- C₆H₁₂O₆ (glucose, a type of sugar)
- NaCl (table salt)

TYPES OF MOLECULES

Molecules are classified according to their composition:

- Diatomic molecule – A diatomic molecule consists of only two atoms. These may be atoms of a single element

(homonuclear) or atoms of different elements (heteronuclear). O₂ is an example of a homonuclear diatomic molecule.

HCl is an example of a heteronuclear diatomic molecule.

- Polyatomic molecule – A polyatomic molecule consists of more than two atoms. Glucose (C₆H₁₂O₆) and water (H₂O)

are examples of polyatomic molecules.

- Macromolecule – A macromolecule is a very large molecule, often consisting of subunits. Proteins and DNA are examples of macromolecules.

It can also be classified base on the chemical bond

- Covalent molecule – Covalent molecules consist of atoms connected by covalent chemical bonds. Covalent molecules consist of nonmetals. In a pure covalent molecule, the atoms have the same electronegativity values.

Examples of pure covalent molecules are H₂ and O₃. In polar covalent molecules, the atoms have slightly different electronegativity values.

Examples of polar covalent molecules are water (H₂O) and hydrochloric acid (HCl).

- Ionic molecule – Ionic molecules consist of both metals and nonmetals (with a few exceptions). The cation (first part of the molecule) and anion (second part of the molecule) have very different electronegativity values. Ionic

molecules display extreme polarity, but usually when someone talks about a polar molecule, they mean a polar covalent molecule. Examples of ionic molecules include salt (NaCl), ammonium acetate ($\text{NH}_4\text{CH}_3\text{CO}_2$ – an ionic compound consisting only of nonmetals), and sodium hydroxide (NaOH).

Difference Between a Molecule and an Ion

A molecule is electrically neutral. The total number of protons and electrons is the same. An ion contains a different number of protons and electrons. If there are more protons, the ion carries a positive charge. If there are more electrons, the ion carries a negative charge. An ion can start as an atom (O^{2-}) or a molecule (H_3O^+). You can tell it apart from a molecule because it always has a + or – superscript for its charge.

Elements are defined by the number of protons they possess. Atoms of an element all have the same number of protons, but they can have different numbers of electrons and neutrons. Changing the ratio of electrons to protons creates ions, while changing the number of neutrons form isotopes. There are 118 known elements. Research is underway to make element 120. When element 120 is made and verified, the periodic table will need to be changed to accommodate it

Elements

Elements are simple substances that cannot be broken down into simpler particles. They are made of only one specific atom. Although elements aren't changed by chemical reactions, new elements may be formed by nuclear reactions. An example of such is carbon C; it is composed of a single carbon atom that is composed of six electrons, six protons, and six neutrons.

Compounds

Compounds are pure substances that are composed of atoms of two or more different elements.

Compounds

can be broken down into separate elements. These substances are made of a fixed ratio of atoms. An example

of a chemical compound is carbon monoxide CO. This compound is made of two elements; oxygen and

carbon. Compounds must have at least two different atoms.

There are four kinds of compounds, depending on how the component atoms are held together:

- molecules held together by covalent bonds.
- ionic compounds held together by ionic bonds.

- intermetallic compounds held together by metallic bonds.
- . Specific complexes held together by coordinate covalent bonds

Compounds vs Elements

The table shows the differences between compounds and elements.

Parameter	Element	Compound
Substances that are made of	two or more	pure substances that are made of
Composition	more different atoms which are the same type of atom	present in definite proportions
They can be broken down into	the	Cannot be separated nor broken
Separation	elements that make the compound	down into simpler particles (through chemical changes)
Existence	Exist independently on their own	Exist in a compound form
The properties of the whole	Properties Possess unique properties	compound is completely different from its constituents (elements)
Representation	Represented by unique symbols	Represented by unique chemical formulas (periodic table symbols)
Carbon dioxide (CO ₂), sodium Helium (He), Chlorine (Cl), Oxygen		Examples chloride (NaCl), water (H ₂ O),

Chemical reaction

A chemical reaction is in which the bonds are broken within reactant molecules, and new bonds are formed within product molecules in order to form a new substance.

Chemical reactions are all around us, from the metabolism of food in our body to how the light we get from the sun is the result of chemical reactions.

We have two major types of changes which occur, these are; Chemical reaction and physical reaction

A burning candle is the best example of physical and chemical change.

Take a candle and light it. As time passes, we can observe that the candle changes to wax. If you cover the candle with a jar, it will extinguish.

In the demonstration, burning of the candle is a chemical change while conversion of the candle to wax is a physical change. In a physical change,

there is basically a change of state of the substance but in the case of a chemical change mostly a new substance is formed in which either energy is given off or absorbed. Thus, we can conclude that chemical changes are accompanied by certain physical changes.

Concepts of Chemical Reactions

- A Chemical Reaction is a process that occurs when two or more molecules interact to form a new product(s).
- Compounds that interact to produce new compounds are called reactants whereas the newly formed compounds are called products.
- Chemical reactions play an integral role in different industries, customs and even in our daily life. They are continuously happening in our general surroundings; for example, rusting of iron, pottery, fermentation of wine and so on.

Concepts of Chemical Reactions (continuation)

- In a chemical reaction, a chemical change must occur which is generally observed with physical changes like precipitation, heat production, colour change etc.
- A reaction can take place between two atoms or ions or molecules, and they form a new bond and no atom is destroyed or created but a new product is formed from reactants.
- The rate of reaction depends on and is affected by factors like pressure, temperature, the concentration of reactants.

Types of Chemical Reactions

The basis for different types of reactions is the product formed, the changes that occur, the reactants involved and so on. Different types of reactions are

Combustion reaction

Decomposition reaction

Neutralization reaction

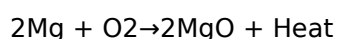
Redox Reaction

Precipitation or Double-Displacement Reaction

Synthesis reaction

1. Combustion Reaction

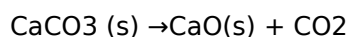
A combustion reaction is a reaction with a combustible material with an oxidizer to give an oxidized product. An oxidizer is a chemical a fuel requires to burn, generally oxygen. Consider the example of combustion of magnesium metal.



Here, 2 magnesium atoms react with a molecule of oxygen producing 2 molecules of the compound magnesium oxide releasing some heat in the process.

2. Decomposition Reaction

A Decomposition reaction is a reaction in which a single component breaks down into multiple products. Certain changes in energy in the environment have to be made like heat, light or electricity breaking bonds of the compound. Consider the example of the decomposition of calcium carbonate giving out CaO (Quick Lime) which is a major component of cement.



3. Neutralization Reaction

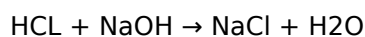
A Neutralization reaction is basically the reaction between an acid and a base giving salt and water as

the products. The water molecule formed is by the combination of OH⁻ ions and H⁺ ions. The overall

pH of the products when a strong acid and a strong base undergo a neutralization reaction will be 7.

Consider the example of the neutralization reaction between Hydrochloric acid and Sodium Hydroxide

giving out sodium chloride(Common Salt) and water.

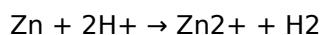


Here, an acid and a base, Hydrochloric acid and Sodium Hydroxide react in a neutralization reaction to

produce Sodium Chloride (Common Salt) and water as the products.

4. Redox Reaction

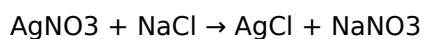
A REDuction-OXidation reaction is a reaction in which there is a transfer of electrons between chemical species. Let us consider the example of an electrochemical cell-like redox reaction between Zinc and Hydrogen.



Here, A Zinc atom reacts with 2 ions of positively charged

5. Precipitation or Double-Displacement Reaction

It is a type of displacement reaction in which two compounds react and consequently, their anions and cations switch places forming two new products. Consider the example of the reaction between silver nitrate and sodium chloride. The products will be silver chloride and sodium nitrate after the double-displacement reaction.

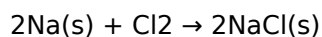


Here, Silver Nitrate and Sodium Chloride undergo a double displacement reaction.

Wherein Silver replaces Sodium in Sodium Chloride and Sodium joins with Nitrate becoming Sodium Nitrate along with the Silver Chloride as the product.

6. Synthesis Reaction

A Synthesis reaction is one of the most basic types of reaction wherein multiple simple compounds combine under certain physical conditions giving out a complex product. The product will always be a compound. Let us consider the Synthesis reaction of sodium chloride with reactants solid sodium and chloride gas.



Here, we have 2 Atoms of solid Sodium reacting with Chlorine gas giving out Sodium Chloride viz. Common Salt as the product.

CHEMICAL EQUATIONS

Chemical equations are symbolic representations of chemical reactions in which the reactants and the products are expressed in terms of their respective chemical formulae.

Chemical equations make use of symbols to represent factors such as the direction of the reaction and the

physical states of the reacting entities. Chemical equations were first formulated by the French chemist

Jean Beguin in the year 1615.

Chemical reactions are represented on paper by chemical equations.

For example, hydrogen gas (H_2) can react (burn) with oxygen gas (O_2) to form water (H_2O). The chemical equation for this reaction is written as: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

This is an example of a , which is a concise way of representing a chemical reaction. The initial substances are called , and the final substances are called .

Representing the Direction of the Chemical Reaction

The reactants and the products (for which the chemical formulae are written in chemical equations) can be separated by one of the following four symbols.

- In order to describe a net forward reaction, the symbol ' $\rightarrow$ ' is used.
- In order to describe a state of chemical equilibrium, the symbol ' $\rightleftharpoons$ ' is used.
- To denote stoichiometric relationships, the '=' symbol is used.
- In order to describe a reaction that occurs in both forward and backward directions, the symbol ' $\rightleftharpoons$ ' is used.

Representing the Physical States of the Reacting Entities

- The symbol (s) describes an entity in the solid state
- The symbol (l) denotes the liquid state of an entity
- The symbol (g) implies that the entity is in the gaseous state.

- The (aq) symbol corresponding to an entity in a chemical equation denotes an aqueous solution of that entity.

Balancing Simple Chemical Equations

When a chemist encounters a new reaction, it does not usually come with a label that shows the balanced chemical equation. Instead, the chemist must identify the reactants and products and then write them in the form of a chemical equation that may or may not be balanced as first written.

Paper 7 of 7 — CHM 101 LECTURE 2 Awojide

Level 100 • First Semester • Handout

CHEMICAL EQUATIONS

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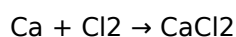
Balancing Simple Chemical Equations

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be balanced as first written.

The fundamental principle behind balancing equations is the law of conservation of mass, which states that matter, meaning physical substances like atoms and molecules, cannot be created or destroyed.

This means there must be the same mass of atoms on both sides of a chemical equation, and therefore the same number of atoms.



REACTANT: 1 atom of Ca and 2 atoms of Cl

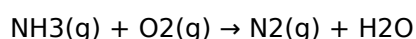
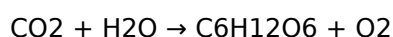
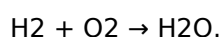
PRODUCT: 1 atom of Ca and 2 atoms of Cl

The equation is balanced because the number of atoms on the reactant side is same as those on the product side

The subscript represents the number of atoms of a given element in each molecule. For example, in 3O_2 , the coefficient is 3 and the subscript is 2.

STEPS FOR BALANCING OF EQUATION:

First, count the atoms on each side. Second, change the coefficient of one of the substances. Third, count the numbers of atoms again and, from there, repeat steps two and three until you've balanced the equation.



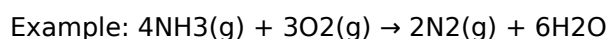
STOICHIOMETRY

Stoichiometry is the quantitative relation between the number of moles (and therefore mass) of various products and reactants in a chemical reaction. Chemical reactions must be balanced or, in other words, must have the same number of various atoms in the products as in the reactants.

Mass Relations from Equation

The relative numbers of reactant and product molecule are indicated by the coefficient of a balance chemical equations.

Using molar masses, we can compare the relative masses of reactants and products in a chemical equation.



4 moles of NH_3 reacts with 3 moles of O_2 to form

2moles of N₂ and 6 moles of H₂O.

Assuming 10g of NH₃ was consumed in the reaction.

□□□□ 10

No of mole = = = 0.588mol of NH₃

□.□.□ 17

From the mole ratio the moles of O₂ consumed as well as N and H₂O produced can be estimated.

4mole of NH₃ → 3 moles of O₂

0.588 mole of NH₃ → x moles of O₂

4=3

0.580 = X

3 □ 0.588

X= = 0.441 mol of O₂

4

A solution containing 2.00g of Hg(NO₃)₂ was added to a solution containing Na₂S. calculate the mass of products formed according to the reaction.

Hg(NO3)2(aq) + Na2S(aq) -> HgS(s) + 2 NaNO3(aq)

1 mole of Hg(NO₃)₂ gives 1 mole of HgS

200 (14 + 16 x 3)₂ → 200 + 32

324.6 → 232

2 →x

324.6 x X = 232 x 2

324.6 x X = 464

464

X= = 1.43 □ □ □

324.6

ASSIGNMENT

In a Rocket motor fueled with butane, C₄H₁₀, how many kilograms of liquid oxygen should be provided with each kilogram of butane to provide complete combustion.

Example 5: 2C5H12OH + 15O2 -> 10CO2 + 12H2O

- How many moles of O₂ are needed for the combustion of 1 mole alkanol?
- How many moles of H₂O are formed for each mole of O₂ consumed?
- How many grams of CO₂ are produced for each mole of alkanol burned
- How many grams of CO₂ are produced for each gram of alkanol burned

Mass concentration.

□□□□ (□)

Mass conc = the unit is g/dm³

□□□□□□ □□³

Example 6: what is the concentration of 4 g of NaOH dissolved in 100 ml of H₂O

□□□□ □

□□□□ () *Convert 100ml to dm³

□□□ □□

100

= 0.1dm³

1000

4

= 40g/dm³

0.1

Molar Concentration

□□□□ □□□□.

Molar conc. = mol/dm³

□.□.□

The concentration of NaOH in mole/dm³ when its concentration in g/dm³ is 40g/dm³

□□□□ □□□□. 40 40

Molar conc = = = = 1 mol/dm³

□.□.□ 23+16+1 40

You are provided with anhydrous Na₂CO₃ of 0.52g which was dissolved in 100ml of distilled water. 25ml of the stock solution of Na₂CO₃ prepared was titrated against HCL to arrive at an average titre value of 20cm³. Calculate

- The number of mole of Na₂CO₃ in the stock solution
- The number of moles of Na₂CO₃ in the 25cm³
- The molarity of the HCl
- The molarity of the Na₂CO₃